HUJI · 769001 · DYNAMICS AND CODING IN THE BRAIN

# Week 02

## The Autapse: Fixed Points, Stability and Bistability

autapse · fixed point · stability · logistic function · bifurcation · bistability · determinant · eigenbasis · mutual inhibition · tanh · E-I network · oscillations

**What this week contains.** Part A: the autapse, from the book ([LMLF §2.3.1]); it will be aligned with the lecture once its content is known. Part B: Bonus hour 2, linear algebra as geometry (determinant, eigenvectors, decoupling two inhibitory neurons, eigenvector shortcuts). Part C: TA session 2, exam 2015 Q2 (one excitatory population with tanh; an E–I pair: trace–determinant regimes and oscillations). Mistakes in the bonus-hour and TA notes are corrected in pitfall boxes. Practice problems and cheat sheets close the week.

CONTENTS

PART A · LECTURE 2 (BOOK §2.3.1)

1 Where we are 2
2 The autapse 2
3 The linear autapse 3
4 Fixed points 4
5 Stability 5
6 The full solution 7
7 The geometric view: the vector field 8
8 Connection to Week 1: it is the eigenvalue 8
9 A limitation of the linear model 9
10 The non-linear autapse: a saturating activation function 9

PART B · BONUS HOUR 2

11 Three ways to look at a matrix 19
12 A matrix moves the plane 19
13 The determinant: how much area changes 20
14 Eigenvectors: the directions a matrix only stretches 21
15 From the network to $\dot{\delta r} = M\delta r$ 21
16 Two inhibitory neurons, solved by decoupling 22
17 Shortcuts for eigenvectors 23

PART C · TA SESSION 2

18 The model and the function tanh 25
19 Question 1.1: one excitatory population 26
20 Question 1.2: an excitatory and an inhibitory population 28

WEEK 02

21 Practice problems with full solutions 31
22 Cheat sheet 34

PART A · LECTURE 2 (BOOK §2.3.1)

# The Autapse

SECTION 01

## Where we are

Week 1 ended with the rate model of a network of $N$ neurons [LMLF eq. 2.8-2.9]:

$$\tau \dot{r}_i = -r_i + g(h_i), \qquad h_i = \sum_{j=1}^{N} J_{ij} r_j + h_i^{\text{ext}}.$$

- $r_i$ is the firing rate of neuron $i$; $\tau$ is its time constant ($\sim$10 ms).
- $h_i$ is its total input: what it hears from the other neurons ($J_{ij} r_j$, synapse *from $j$ to $i$*) plus external input $h_i^{\text{ext}}$.
- $g$ is the activation function: it turns input into a rate.

**THE PLAN FOR THIS WEEK**

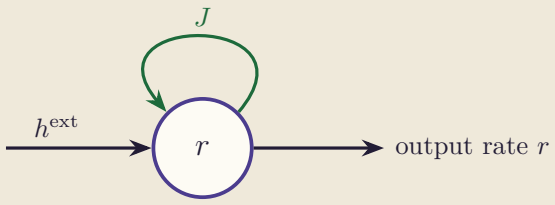
Before analysing big networks, study the smallest possible network: one neuron. If it talks to itself, it already has *feedback*, and feedback is what makes networks interesting. This week we use it to introduce two tools we will use all course long: **fixed points** and their **stability**. We do it twice: first with a linear activation function, where everything can be solved exactly, then with a saturating one (§10), where the same neuron becomes **bistable** and can store a memory.

SECTION 02

## The autapse

**DEF · AUTAPSE**

An autapse is a synapse that a neuron makes onto itself: its axon connects back to its own dendrites. A neuron with an autapse is *the simplest possible network*: one neuron, one connection, of strength $J$ [LMLF 2.3.1].




The autapse (after [LMLF Fig. 2.6]). The neuron receives an external input $h^{\text{ext}}$, and its own output $r$ comes back to it through the synapse of strength $J$.

With a single neuron, the network equations lose their indices [LMLF eq. 2.11-2.12]:

**AUTAPSE DYNAMICS**

$$\tau \dot{r} = -r + g(h), \qquad h = Jr + h^{\text{ext}}.$$

**DEF · EXCITATORY AUTAPSE**

We focus on excitatory autapses, $J > 0$: the more the neuron fires, the more input it gives itself. This is positive feedback.

**EXAMPLE · ONE INPUT VALUE, TWO SOURCES**

$J = 0.5$, $h^{\text{ext}} = 1$, current rate $r = 4$: total input $h = 0.5 \cdot 4 + 1 = 3$. Two of the three units of input come from the neuron itself.

SECTION 03

# The linear autapse

**DEF · LINEAR ACTIVATION FUNCTION**

The linear (identity) activation function is $g(h) = h$: the rate equals the input, with no threshold and no saturation [LMLF eq. 2.4].

**PITFALL · WHY LINEAR IS A SIMPLIFICATION**

A real neuron cannot fire at a negative rate, and it cannot fire faster than a few hundred Hz. The linear $g$ ignores both limits. We use it because the resulting equation is linear, so we can solve it and understand it completely. It is a good approximation when the rate changes only a little.

With $g(h) = h$ the autapse equation becomes

$$\tau \dot{r} = -r + Jr + h^{\text{ext}} \qquad \Longrightarrow \qquad \boxed{\tau \dot{r} = -(1 - J)\, r + h^{\text{ext}}}$$

**TWO FORCES ON $r$**

- $-r$: the leak. Left alone, the rate decays back to zero (Week 1).
- $+Jr$: the feedback. The neuron excites itself in proportion to its own rate.

The whole story of this week is the contest between leak and feedback: who wins depends on whether $J < 1$ or $J > 1$.

This is exactly the equation $\dot{n} = an + I$ from the Week 1 bonus hour, with $a = -(1 - J)/\tau$ and constant input $I = h^{\text{ext}}/\tau$. We could solve it right away (we do so in §6). But first we introduce an approach that works *even when the equation cannot be solved*: look for the fixed points and ask whether they are stable.

SECTION 04

# Fixed points

**DEF · FIXED POINT**

A fixed point $r^*$ is a value of $r$ where the dynamics stand still: $\dot{r} = 0$. If the system starts exactly at $r^*$, it stays there forever [LMLF 2.3.1].

**Finding it.** Set the right-hand side to zero:

$$0 = -(1 - J)\, r^* + h^{\text{ext}} \qquad \Longrightarrow \qquad \boxed{r^* = \frac{h^{\text{ext}}}{1 - J}} \qquad (J \neq 1).$$

**EXAMPLE · FIXED POINTS FOR $h^{\text{ext}} = 1$**

| $J$ | $r^* = 1/(1 - J)$ | COMMENT |
|---|---|---|
| 0 | 1 | no feedback: the rate just copies the input |
| 0.5 | 2 | feedback doubles the response |
| 0.9 | 10 | strong feedback: ten times the input |
| 0.99 | 100 | close to $J = 1$ the fixed point runs off |
| 1.5 | $-2$ | negative! (see below) |

**FEEDBACK AMPLIFIES**

For $0 < J < 1$, $r^* = h^{\text{ext}}/(1 - J) > h^{\text{ext}}$: the autapse amplifies its input by the factor $1/(1 - J)$. Why: the input raises $r$, which raises the feedback $Jr$, which raises $r$ further, and so on. The extra contributions add up to $h^{\text{ext}}(1 + J + J^2 + \dots) = h^{\text{ext}}/(1 - J)$, a geometric series that converges only if $J < 1$.

**PITFALL · A FIXED POINT IS NOT AUTOMATICALLY WHERE THE SYSTEM GOES**

For $J = 1.5$ the formula gives $r^* = -2$. It is a perfectly valid solution of $\dot{r} = 0$, but (as we will see) the system runs *away* from it. Also, a negative rate shows that the linear model has left the range where it describes a real neuron. Finding a fixed point is only half the job; the other half is its stability.

**PITFALL · THE BORDER CASE $J = 1$**

For $J = 1$ the leak and the feedback cancel exactly: $\tau\dot{r} = h^{\text{ext}}$. If $h^{\text{ext}} \neq 0$ there is *no* fixed point ($r$ grows or shrinks steadily). If $h^{\text{ext}} = 0$ then $\dot{r} = 0$ everywhere: *every* value of $r$ is a fixed point. The book returns to this special case later (§4.2); this week we assume $J \neq 1$.

SECTION 05

# Stability

## > 5.1 The idea

**DEF · STABLE AND UNSTABLE FIXED POINTS**

Start the system *near* $r^*$ instead of exactly on it.

- If the dynamics bring $r$ ever closer to $r^*$, the fixed point is stable.
- If the dynamics push $r$ away from $r^*$, the fixed point is unstable.



stable: a ball in a valley unstable: a ball on a hilltop

Both balls are at rest (fixed points). A small nudge brings the left ball back but sends the right one rolling away.

**WHY STABILITY MATTERS**

In the real brain the rate is never exactly at $r^*$: there is always noise. An unstable fixed point is never seen in practice, because the tiniest disturbance grows. Only stable fixed points describe states the network actually sits in.

## > 5.2 Perturbation analysis, step by step

To decide stability, follow a small deviation from the fixed point [LMLF eq. 2.14-2.15].

**DEF · PERTURBATION**

The perturbation is the distance from the fixed point: $\delta r(t) = r(t) - r^*$. Equivalently $r = r^* + \delta r$.

1. **Substitute** $r = r^* + \delta r$ into $\tau \dot{r} = -r + Jr + h^{\text{ext}}$:

$$\tau \frac{d}{dt}(r^* + \delta r) = -(r^* + \delta r) + J(r^* + \delta r) + h^{\text{ext}}.$$

2. $r^*$ **is a number, not a function of time**, so $\frac{d}{dt} r^* = 0$ and the left side is $\tau\, \dot{\delta r}$.

3. **Regroup the right side** into the part with $r^*$ and the part with $\delta r$:

$$\tau\, \dot{\delta r} = \underbrace{\big(-r^* + Jr^* + h^{\text{ext}}\big)}_{=0 \text{ (definition of } r^*)} + \big(-\delta r + J\delta r\big).$$

4. **The first bracket vanishes** because $r^*$ is a fixed point. What remains is

$$\boxed{\tau\, \dot{\delta r} = -(1 - J)\, \delta r}$$

WHAT HAPPENED

The external input $h^{\text{ext}}$ has disappeared. It only decides *where* the fixed point is; whether the fixed point is stable depends only on $J$.

## > 5.3 Reading off stability from the sign

**Case** $J < 1$, so $1 - J > 0$:

- if $\delta r > 0$ (rate above the fixed point) then $\dot{\delta r} < 0$: it comes down;
- if $\delta r < 0$ (rate below) then $\dot{\delta r} > 0$: it goes up.

Either way the perturbation shrinks. The fixed point is stable.

**Case** $J > 1$, so $1 - J < 0$: now $\dot{\delta r}$ has the *same* sign as $\delta r$. Above the fixed point the rate rises further; below it, it falls further. The fixed point is unstable.

STABILITY OF THE LINEAR AUTAPSE

$$J < 1 \;\Rightarrow\; \text{stable} \qquad\qquad J > 1 \;\Rightarrow\; \text{unstable}$$

$J = 1$ is the border between two qualitatively different behaviours.

LEAK VS. FEEDBACK AGAIN

$J < 1$: when the rate goes up by $\delta r$, the leak pulls it down by $\delta r$ but the feedback pushes it up only by $J\delta r < \delta r$. Leak wins, the deviation dies. $J > 1$: the feedback push $J\delta r$ beats the leak, the deviation feeds itself.

## > 5.4 How fast? Solving for $\delta r(t)$

$\tau\dot{\delta r} = -(1 - J)\delta r$ is the Week 1 equation $\dot{n} = an$ with $a = -(1 - J)/\tau$, so [LMLF eq. 2.16]

PERTURBATION OVER TIME

$$\delta r(t) = \delta r(0)\, e^{-(1-J)t/\tau}.$$

- $J < 1$: exponential **decay** to 0, i.e. $r(t) \to r^*$.
- $J > 1$: exponential **growth**; $r$ diverges to $+\infty$ or $-\infty$, depending on the sign of $\delta r(0)$.

EXAMPLE · HOW LONG A PERTURBATION LASTS, $\tau = 10\,\text{MS}$

The perturbation shrinks (or grows) by a factor $e$ every $\tau/|1 - J|$:

| $J$ | $\tau/\lvert 1 - J\rvert$ | BEHAVIOUR |
|---|---|---|
| 0 | 10 ms | decays (just the leak) |
| 0.5 | 20 ms | decays, twice as slowly |
| 0.9 | 100 ms | decays, ten times as slowly |
| 1.5 | 20 ms | grows by $e$ every 20 ms |

The closer $J$ is to 1, the slower everything happens: stronger feedback lets the neuron hold on to a deviation for much longer than its own $\tau$.

SECTION 06

# The full solution

Using the Week 1 result for $\dot{n} = an + I$ with constant input (or simply $r = r^* + \delta r$ with $\delta r(t)$ from above):

LINEAR AUTAPSE, SOLVED

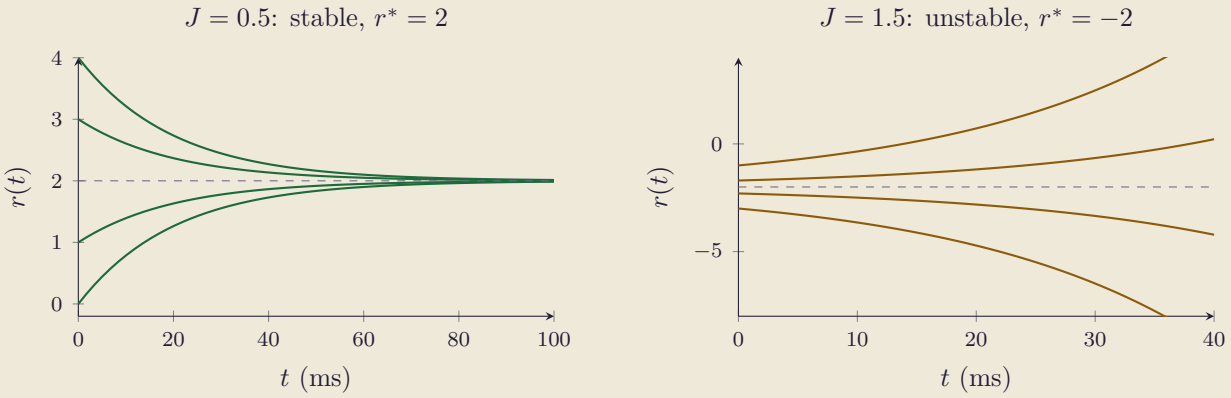
$$r(t) = r^* + \big(r(0) - r^*\big)\, e^{-(1-J)t/\tau}, \qquad r^* = \frac{h^{\text{ext}}}{1-J}.$$

**Check.** At $t = 0$: $r(0) = r^* + r(0) - r^* = r(0)$ ✓. Differentiate: $\tau\dot{r} = -(1-J)(r(0)-r^*)e^{-(1-J)t/\tau} = -(1-J)(r - r^*) = -(1-J)r + (1-J)r^* = -(1-J)r + h^{\text{ext}}$. ✓

EXAMPLE · STARTING FROM SILENCE, $r(0) = 0$, $h^{\text{ext}} = 1$, $\tau = 10\,\text{MS}$

| $t$ (ms) | 0 | 10 | 20 | 40 | 60 | 100 |
|---|---|---|---|---|---|---|
| $J = 0.5$: $2(1 - e^{-t/20})$ | 0 | 0.79 | 1.26 | 1.73 | 1.90 | 1.99 |
| $J = 0.9$: $10(1 - e^{-t/100})$ | 0 | 0.95 | 1.81 | 3.30 | 4.51 | 6.32 |
| $J = 1.5$: $-2 + 2e^{t/20}$ | 0 | 1.30 | 3.44 | 12.8 | 38.2 | 295 |

$J = 0.5$ settles at 2 within ∼60 ms. $J = 0.9$ heads for 10 but needs hundreds of ms. $J = 1.5$ starts *above* its fixed point $-2$, so it is pushed further and further up, without limit.




$r(t)$ from four starting points ($\tau = 10$ ms, $h^{\text{ext}} = 1$). Left: every trajectory converges to the dashed fixed point. Right: every trajectory that does not start exactly on the fixed point is driven away from it.

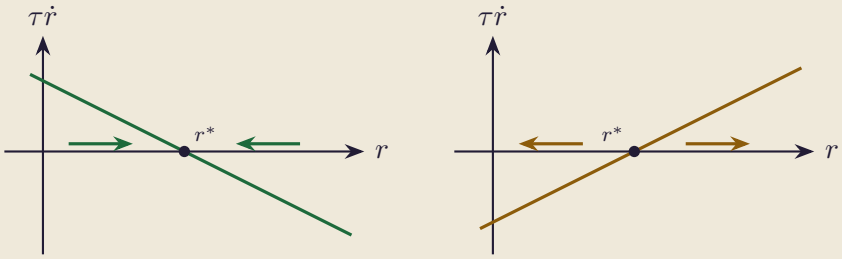
SECTION 07

# The geometric view: the vector field

We can see stability *without solving anything*, by plotting $\dot{r}$ as a function of $r$ (the idea from the Week 1 bonus hour).

**DEF · VECTOR FIELD ON A LINE**

The equation $\tau\dot{r} = F(r)$ assigns a velocity $\dot{r}$ to every value of $r$: a vector field on the line [LMLF 2.3.1]. Draw $\dot{r}$ against $r$ and put arrows on the $r$-axis: right where $\dot{r} > 0$, left where $\dot{r} < 0$. The fixed points are where the graph crosses zero.




$J < 1$: slope $-(1-J) < 0$
arrows point *in*: stable

$J > 1$: slope $J - 1 > 0$
arrows point *out*: unstable

$\tau\dot{r} = -(1-J)r + h^{\text{ext}}$ is a straight line in $r$ that crosses zero at $r^*$. Its slope decides everything.

**STABILITY FROM THE SLOPE**

A fixed point is stable if the graph of $\dot{r}$ against $r$ crosses zero going **down** (negative slope), and unstable if it crosses going **up** (positive slope). For the linear autapse the slope is $-(1-J)/\tau$: negative exactly when $J < 1$.

**WHY THIS IS POWERFUL**

This picture needs only the *sign* of $\dot{r}$ around $r^*$. It works for equations we cannot solve, which will be the usual case once $g$ is not linear.

SECTION 08

# Connection to Week 1: it is the eigenvalue

In the TA session of Week 1 we saw that $\dot{\vec{x}} = M\vec{x}$ converges iff all eigenvalues of $M$ are negative. The perturbation equation

$$\dot{\delta r} = \underbrace{-\frac{1-J}{\tau}}_{\text{a } 1\times 1 \text{ “matrix”}} \delta r$$

is the one-dimensional case: its only eigenvalue is $-(1-J)/\tau$, negative exactly when $J < 1$. Stability of a fixed point = the sign of the eigenvalue of the perturbation dynamics. For networks

of many neurons the same recipe gives a matrix, and Week 1's eigenvalue tools take over.

SECTION 09

# A limitation of the linear model

PITFALL · DIVERGENCE IS NOT BIOLOGY

For $J > 1$ the linear autapse fires at a rate that grows without bound (or goes to $-\infty$). Real neurons *saturate*: their rate cannot exceed a few hundred Hz, and it cannot go below zero. This is a limitation of the linear $g$, not of the autapse itself. It is the reason the book next replaces $g(h) = h$ by a saturating activation function [LMLF 2.3.1].

SECTION 10

# The non-linear autapse: a saturating activation function

The rest of this week keeps the *same* autapse, $\tau\dot{r} = -r + g(Jr + h^{\text{ext}})$, and changes only $g$. The result is surprising: a single neuron with strong feedback no longer explodes. Instead it gets two stable states, and it can use them to remember a bit of information.

Two misreadings to clear up first:

- **The non-linearity does not make the analysis hopeless.** We can no longer write down $r(t)$, but fixed points and their stability can still be found, with a picture.
- **Stability is no longer a property of the whole system.** For the linear autapse, $J$ alone decided stability. Now each fixed point has *its own* stability, and one neuron can have a stable and an unstable fixed point side by side.

## > 10.1 The logistic function

DEF · LOGISTIC ACTIVATION FUNCTION

The logistic function [LMLF eq. 2.2] is

$$g(x) = \frac{1}{1 + e^{-\beta(x-\theta)}}.$$

It is sigmoidal (S-shaped): it increases monotonically, it is bounded below by 0 (no negative rates) and above by 1 (saturation). The rate $r$ is therefore *normalised*: $r = 1$ means "firing at the neuron's maximal rate", not 1 Hz [LMLF 2.2].

It has two parameters:

**DEF · THRESHOLD $\theta$ AND GAIN $\beta$**

- The threshold $\theta$ is the input at which the rate is exactly half its maximum: $g(\theta) = 1/(1+e^0) = 0.5$. Changing $\theta$ slides the curve left or right.
- The gain $\beta$ sets how steep the curve is. Its maximal slope, reached at $x = \theta$, is $\beta/4$ (derived below).

**EXAMPLE · THE LOGISTIC FUNCTION WITH $\theta = 0$, $\beta = 20$ (THE BOOK'S VALUES)**

| $x$ | $-0.3$ | $-0.2$ | $-0.1$ | $-0.05$ | $0$ | $0.05$ | $0.1$ | $0.2$ | $0.3$ |
|---|---|---|---|---|---|---|---|---|---|
| $g(x)$ | 0.002 | 0.018 | 0.119 | 0.269 | 0.5 | 0.731 | 0.881 | 0.982 | 0.998 |
| $g'(x)$ | 0.05 | 0.35 | 2.1 | 3.9 | 5 | 3.9 | 2.1 | 0.35 | 0.05 |

Almost all of the change happens for $x$ between $-0.2$ and $0.2$. Outside that window the neuron is either nearly silent or nearly saturated, and the slope is close to 0.

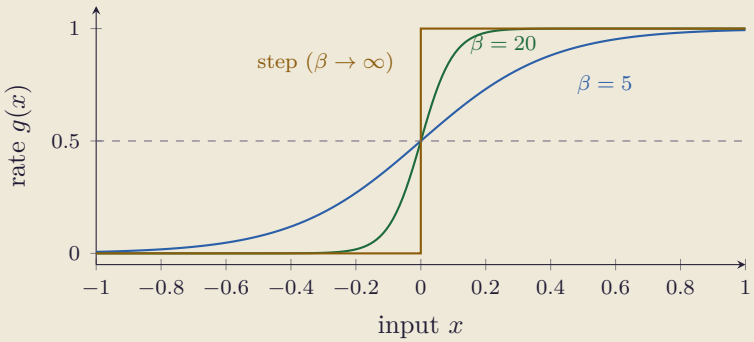
**The slope of the logistic function**, step by step (we need it for stability). Write $u = e^{-\beta(x-\theta)}$, so $g = 1/(1+u)$ and $du/dx = -\beta u$:

$$g'(x) = -\frac{1}{(1+u)^2} \cdot (-\beta u) = \beta \, \frac{1}{1+u} \cdot \frac{u}{1+u} = \beta \, g(x)(1 - g(x)).$$

The product $g(1-g)$ is largest when $g = \frac{1}{2}$, i.e. at $x = \theta$, where it equals $\frac{1}{4}$. So

**SLOPE OF THE LOGISTIC**

$$g'(x) = \beta \, g(x) \, (1 - g(x)), \qquad \max_x g'(x) = g'(\theta) = \frac{\beta}{4}.$$




The logistic function with $\theta = 0$ (after [LMLF Fig. 2.2a]). All curves pass through $(\theta, 0.5)$. A larger gain $\beta$ gives a steeper curve. In the limit $\beta \to \infty$ it becomes a step: 0 below threshold, 1 above.

## > 10.2 The fixed-point equation

Setting $\dot{r} = 0$ in $\tau \dot{r} = -r + g(Jr + h^{\text{ext}})$ gives [LMLF eq. 2.17]

**FIXED POINTS OF THE NON-LINEAR AUTAPSE**

$$r^* = g(Jr^* + h^{\text{ext}}).$$

**PITFALL · YOU CANNOT SOLVE THIS FOR $r^*$ WITH ALGEBRA**

$r^*$ appears on both sides, once inside an exponential. There is no formula like $r^* = h^{\text{ext}}/(1-J)$ any more. Do not try to isolate $r^*$. Instead, read the equation as a question about two curves.

**The geometric method**, step by step:

1. **Left side**: plot $y = r$ against $r$. This is the identity line (slope 1).
2. **Right side**: plot $y = g(Jr + h^{\text{ext}})$ against $r$. As a function of $r$ this is *again a logistic function*, with new parameters. Substitute $x = Jr + h^{\text{ext}}$:

$$\beta(Jr + h^{\text{ext}} - \theta) = (\beta J)\Big(r - \frac{\theta - h^{\text{ext}}}{J}\Big),$$

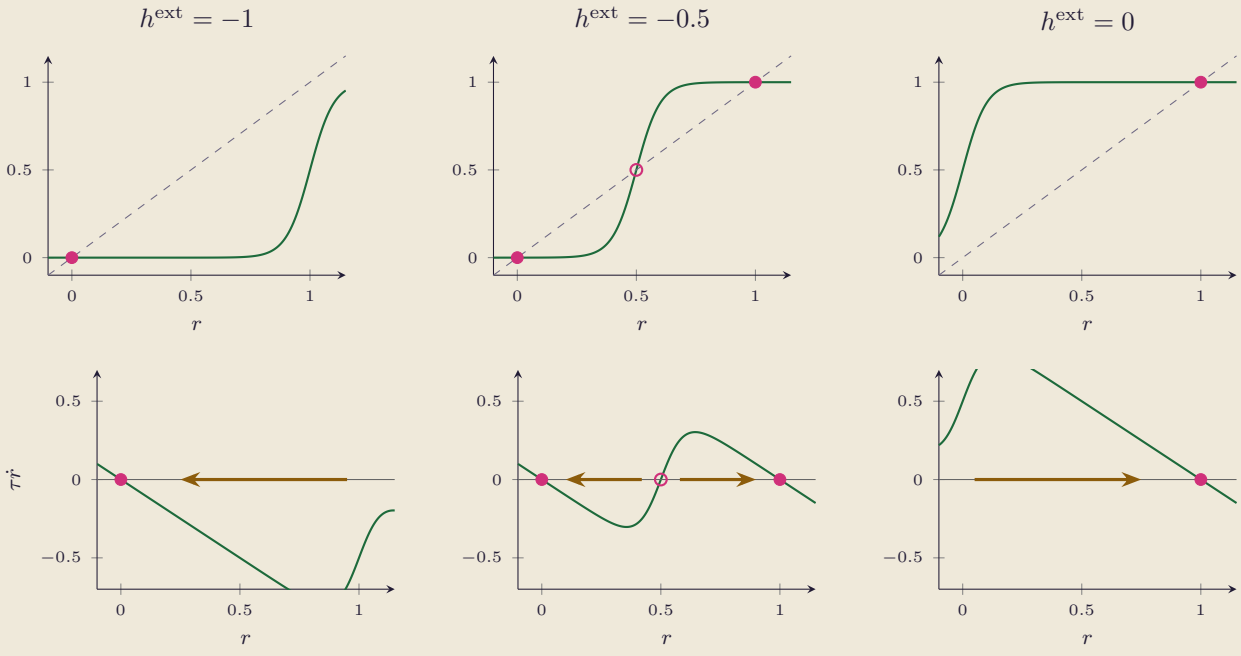
   so it has effective gain $\beta J$ and effective threshold $(\theta - h^{\text{ext}})/J$ [LMLF 2.3.1].
3. **Every intersection** of the two curves satisfies $r = g(Jr + h^{\text{ext}})$, so it is a fixed point. Counting intersections counts fixed points.

**WHAT $h^{\text{ext}}$ DOES IN THE PICTURE**

$h^{\text{ext}}$ appears only in the effective threshold $(\theta - h^{\text{ext}})/J$. Raising $h^{\text{ext}}$ lowers the threshold, so it slides the S-curve to the left, along the $r$ axis. The neuron then reaches half its maximal rate already at a smaller $r$, because the external input supplies part of the drive.

**PITFALL · LEFT, NOT RIGHT**

The book writes that increasing $h^{\text{ext}}$ "results in a shift of the logistic function to the right". As a function of $r$, the curve moves to the *left*: its midpoint $(\theta - h^{\text{ext}})/J$ decreases when $h^{\text{ext}}$ grows. Check with the book's own numbers ($\theta = 0$, $J = 1$): the midpoint is at $r = 1$ for $h^{\text{ext}} = -1$, at $r = 0.5$ for $h^{\text{ext}} = -0.5$ and at $r = 0$ for $h^{\text{ext}} = 0$. The figure below (and [LMLF Fig. 2.7]) shows exactly this leftward slide.



Redrawn after [LMLF Fig. 2.7]; $\theta = 0$, $\beta = 20$, $J = 1$. **Top**: the S-curve $g(r + h^{\text{ext}})$ (green) and the identity line (dashed); the intersections are the fixed points. **Bottom**: the vector field $\tau\dot{r} = -r + g(r + h^{\text{ext}})$ with the direction of motion on the $r$ axis (arrows). Filled dots: stable fixed points; open dot: unstable. Left: one low-activity fixed point. Middle: three fixed points. Right: one high-activity fixed point.

**EXAMPLE · FINDING THE FIXED POINTS FOR $h^{\text{ext}} = -0.5$ BY HAND**

The equation is $r^* = g(r^* - 0.5)$ with $g(x) = 1/(1 + e^{-20x})$.

- **Middle**: try $r^* = 0.5$. Then $g(0) = 0.5$ ✓. It is a fixed point exactly.
- **Low**: try $r^* \approx 0$. Then $g(-0.5) = 1/(1 + e^{10}) \approx 4.5 \times 10^{-5}$. So $r^* \approx 0.000045$: the neuron is practically silent.
- **High**: try $r^* \approx 1$. Then $g(0.5) = 1/(1 + e^{-10}) \approx 1 - 4.5 \times 10^{-5}$. So $r^* \approx 0.99995$: practically saturated.

Each guess is then improved by plugging the result back in, but the change is in the fifth decimal place.

## > 10.3 Stability from the vector field

We read stability off the bottom row, exactly as in the linear case: arrows pointing *in* mean stable, arrows pointing *out* mean unstable.

**EXAMPLE · THE SIGN OF $\tau\dot{r}$ FOR $h^{\text{ext}} = -0.5$**

| $r$ | 0 | 0.1 | 0.25 | 0.4 | **0.5** | 0.6 | 0.75 | 0.9 | 1 |
|---|---|---|---|---|---|---|---|---|---|
| $g(r - 0.5)$ | 0.000 | 0.000 | 0.007 | 0.119 | 0.5 | 0.881 | 0.993 | 1.000 | 1.000 |
| $\tau\dot{r}$ | $\approx 0$ | $-0.10$ | $-0.24$ | $-0.28$ | **0** | $+0.28$ | $+0.24$ | $+0.10$ | $\approx 0$ |
| motion | • | ← | ← | ← | ∘ | → | → | → | • |

Between 0 and 0.5 the rate falls, towards 0. Between 0.5 and 1 it rises, towards 1. So 0 and 1 attract, and 0.5 repels: it is the watershed that decides which way the neuron goes.

For a single fixed point (left and right panels) the arrows point towards $r^*$ from *every* starting value. For three fixed points, the outer ones attract only the starting values on their own side of 0.5. The book separates these two situations [LMLF 2.3.1]:

**DEF · GLOBAL AND LOCAL STABILITY**

- A fixed point is globally stable if the dynamics reach it from every starting value $r \neq r^*$. Example: the only fixed point for $h^{\text{ext}} = -1$.
- It is locally stable if the dynamics reach it from every starting value close enough to it, but not necessarily from far away. Example: $r^* \approx 0$ for $h^{\text{ext}} = -0.5$. Starting at $r = 0.3$ leads there, starting at $r = 0.7$ does not.

Every globally stable fixed point is locally stable; the converse is false.

## > 10.4 Stability by linearisation

The picture is convincing, but we also want a number, as in the linear case. The trick is to make the equation linear near the fixed point.

**DEF · FIRST-ORDER TAYLOR EXPANSION**

Near a point $a$, a smooth function is approximately a straight line: $f(a+\varepsilon) \approx f(a) + f'(a)\,\varepsilon$ for small $\varepsilon$. **Example**: $g(x) = 1/(1+e^{-20x})$ at $a = 0.1$, with $\varepsilon = 0.01$. Exact: $g(0.11) = 0.9002$. Approximation: $g(0.1) + g'(0.1) \cdot 0.01 = 0.8808 + 2.1 \cdot 0.01 = 0.9018$. Close, because $\varepsilon$ is small.

Follow the same recipe as for the linear autapse [LMLF eq. 2.18-2.22]:

1. **Substitute** $r = r^* + \delta r$:

$$\tau \frac{d}{dt}(r^* + \delta r) = -(r^* + \delta r) + g\big(J(r^* + \delta r) + h^{\text{ext}}\big).$$

2. **Name the input at the fixed point** $h^* = Jr^* + h^{\text{ext}}$. The argument of $g$ is then $h^* + J\delta r$.

3. **Taylor-expand** $g$ around $h^*$ with the small step $\varepsilon = J\delta r$:

$$g(h^* + J\delta r) \approx g(h^*) + g'(h^*)\, J\delta r.$$

4. **Regroup** (and use $\frac{d}{dt} r^* = 0$):

$$\tau\, \dot{\delta r} = \underbrace{\big(-r^* + g(h^*)\big)}_{=0 \text{ (fixed point)}} - \delta r + J g'(h^*)\, \delta r.$$

5. **Result**, with the number $\lambda = J g'(h^*) = J g'(Jr^* + h^{\text{ext}})$:

$$\boxed{\tau\, \dot{\delta r} = -(1-\lambda)\, \delta r} \qquad \Longrightarrow \qquad \delta r(t) = \delta r(0)\, e^{-(1-\lambda)t/\tau}.$$

**LOCAL STABILITY OF A FIXED POINT OF THE AUTAPSE**

$$\lambda = J\, g'(Jr^* + h^{\text{ext}}): \qquad \lambda < 1 \;\Rightarrow\; \text{stable}, \qquad \lambda > 1 \;\Rightarrow\; \text{unstable}.$$

**THE SAME EQUATION AS BEFORE, WITH $J$ REPLACED BY $\lambda$**

Compare with $\tau\dot{\delta r} = -(1-J)\delta r$ for the linear autapse. Near a fixed point, the non-linear neuron behaves like a *linear* autapse whose feedback strength is the effective feedback $\lambda = Jg'(h^*)$: the synaptic weight times how sensitive the neuron is at its current operating point. For $g(h) = h$, $g' = 1$ and $\lambda = J$: we recover the old result.

**EXAMPLE · $\lambda$ AT THE THREE FIXED POINTS FOR $h^{\text{ext}} = -0.5$ ($J = 1$, $\beta = 20$, $\tau = 10\,\text{MS}$)**

Use $g' = \beta g(1-g)$ and $g(h^*) = r^*$ at a fixed point, so $\lambda = J\beta\, r^*(1-r^*)$:

| $r^*$ | $\lambda = 20\,r^*(1-r^*)$ | VERDICT | TIME SCALE $\tau/\lvert 1-\lambda\rvert$ |
|---|---|---|---|
| 0.000045 | 0.0009 | stable | $\approx 10$ ms decay |
| 0.5 | 5 | unstable | 2.5 ms growth |
| 0.99995 | 0.0009 | stable | $\approx 10$ ms decay |

At the outer fixed points the neuron is silent or saturated, so the curve is flat ($g' \approx 0$): feedback does nothing and only the leak acts. At 0.5 the curve is steepest: a deviation of 0.01 grows by a factor $e$ every 2.5 ms.

**The geometric version.** $\lambda = Jg'(Jr^* + h^{\text{ext}}) = \frac{d}{dr}\, g(Jr + h^{\text{ext}})\big|_{r^*}$ is the slope of the S-curve at the intersection. The identity line has slope 1. So:

STABILITY AT A GLANCE

In the top-row picture, a fixed point is **stable** if the S-curve crosses the identity line **less steeply** than it (slope $< 1$), and **unstable** if it crosses **more steeply** (slope $> 1$).

Look again at the figure: the single crossings (left, right) and the two outer crossings (middle) are flat; the middle crossing is steep.

## > 10.5 Bifurcations

As $h^{\text{ext}}$ increases smoothly from $-1$ to 0, the number of fixed points jumps: $1 \to 3 \to 1$. This kind of event has a name [LMLF 2.3.1].

DEF · BIFURCATION

A bifurcation is a *qualitative* change in the dynamics (fixed points appear or disappear, or a fixed point changes its stability) caused by a *smooth* change of a parameter. The parameter values where it happens are the bifurcation points.

DEF · SADDLE-NODE BIFURCATION

In a saddle-node bifurcation a stable and an unstable fixed point move towards each other, merge into one, and disappear (or, run backwards, are born together out of nothing). Both transitions of the autapse are of this type. At the moment of merging, the S-curve just touches the identity line: the two curves are tangent.

EXAMPLE · COMPUTING THE TWO BIFURCATION POINTS ($\theta = 0$, $\beta = 20$, $J = 1$)

At a bifurcation point the S-curve is tangent to the identity line, so two conditions hold at once:

1. it is a fixed point: $r^* = g(h^*)$, with $h^* = r^* + h^{\text{ext}}$;
2. the slopes are equal: $\lambda = J\beta\, g(1-g) = 1$, i.e. $20\,r^*(1-r^*) = 1$.

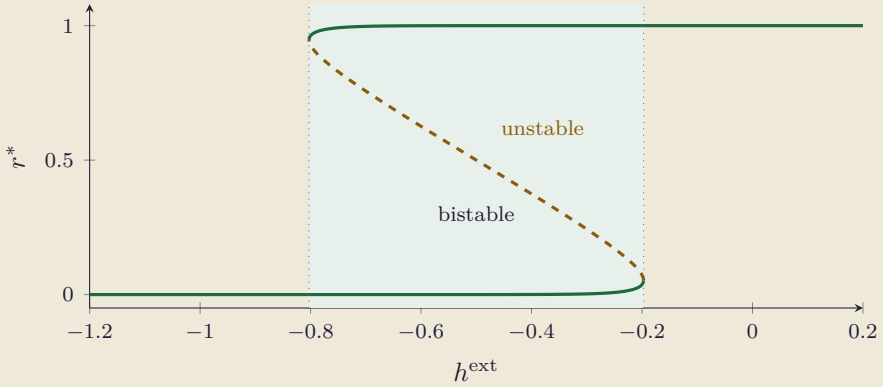
**Step 1.** Condition 2 is a quadratic: $r^{*2} - r^* + 0.05 = 0$, so $r^* = \frac{1}{2}\left(1 \pm \sqrt{0.8}\right)$, i.e. $r^* = 0.947$ or $r^* = 0.053$.

**Step 2.** Invert the logistic to find the input: $g(h^*) = r^*$ gives $h^* = \frac{1}{\beta} \ln \frac{r^*}{1-r^*}$. For $r^* = 0.947$: $h^* = \frac{1}{20} \ln 17.9 = 0.144$. For $r^* = 0.053$: $h^* = -0.144$.
**Step 3.** $h^{\text{ext}} = h^* - Jr^*$: $0.144 - 0.947 = \mathbf{-0.803}$ and $-0.144 - 0.053 = \mathbf{-0.197}$.
So the autapse has three fixed points exactly for $-0.803 < h^{\text{ext}} < -0.197$. The book's examples $h^{\text{ext}} = -1$ (below), $-0.5$ (inside), $0$ (above) fall where they should.

The whole story fits in one picture: plot every fixed point against $h^{\text{ext}}$.

DEF · BIFURCATION DIAGRAM

A bifurcation diagram plots the fixed points (vertical axis) against a parameter (horizontal axis). Stable fixed points are drawn solid, unstable ones dashed.




Bifurcation diagram, redrawn after [LMLF Fig. 2.8] ($\theta = 0$, $\beta = 20$, $J = 1$). Solid: stable fixed points; dashed: unstable. In the shaded band, $-0.803 < h^{\text{ext}} < -0.197$, there are three fixed points. At each edge of the band the dashed branch merges with a solid one: a saddle-node bifurcation.

## > 10.6 Attractors and bistability

DEF · STATE SPACE AND ATTRACTOR

The state space is the set of all possible states of the system; for the autapse it is just the $r$ axis. An attractor is a region of state space to which all nearby trajectories converge [LMLF 2.3.1]. A stable fixed point is the simplest attractor; others appear later in the course.

DEF · BISTABILITY

A system is bistable if it has two stable fixed points (two attractors). Between them there is always an unstable fixed point, which separates the starting values that go to one from those that go to the other. The autapse is bistable for $-0.803 < h^{\text{ext}} < -0.197$ (with $\theta = 0$, $\beta = 20$, $J = 1$).

EXAMPLE · WHERE DOES THE NEURON END UP? ($h^{\text{ext}} = -0.5$, $\tau = 10\,\text{MS}$)

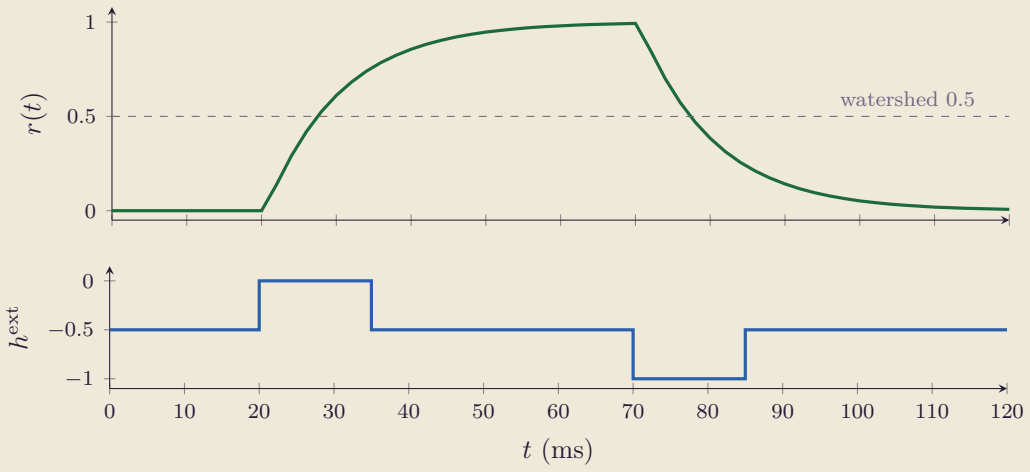
| $t$ (ms) | 0 | 10 | 20 | 30 | 50 | 70 | 100 |
|---|---|---|---|---|---|---|---|
| $r(0) = 0.40$ | 0.40 | 0.157 | 0.058 | 0.021 | 0.003 | 0.000 | 0.000 |
| $r(0) = 0.45$ | 0.45 | 0.194 | 0.072 | 0.026 | 0.004 | 0.001 | 0.000 |
| $r(0) = 0.55$ | 0.55 | 0.806 | 0.928 | 0.974 | 0.996 | 0.999 | 1.000 |
| $r(0) = 0.60$ | 0.60 | 0.843 | 0.942 | 0.979 | 0.997 | 1.000 | 1.000 |

(Numerical solutions.) Starting values only 0.1 apart end in opposite states. The final state is decided entirely by which side of 0.5 the neuron starts on.

## > 10.7 Memory: switching with a brief input

Why care about two stable states? Because a bistable neuron can store one bit [LMLF 2.3.1]. Left alone in the bistable range, it stays forever in whichever state it is in. To write a bit, change $h^{\text{ext}}$ *briefly*:

1. The neuron rests in the low state, $h^{\text{ext}} = -0.5$.
2. A brief pulse raises $h^{\text{ext}}$ to 0, *above* the bifurcation point $-0.197$. Now the low fixed point no longer exists; the only attractor is the high one, and $r$ starts climbing.
3. The pulse ends: $h^{\text{ext}}$ returns to $-0.5$. The low state exists again, but if $r$ has already passed the watershed 0.5, it keeps going to the high state.
4. The neuron stays high indefinitely, although the input that put it there is long gone. That is memory.
5. To erase, pulse $h^{\text{ext}}$ *below* $-0.803$ (here to $-1$): the high state disappears and the neuron falls back to the low one, where it stays.




State transitions, simulated after [LMLF Fig. 2.9] ($\theta = 0$, $\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$). A 15 ms pulse to $h^{\text{ext}} = 0$ switches the neuron on (it has reached $0.76 > 0.5$ when the pulse ends); a 15 ms pulse to $h^{\text{ext}} = -1$ switches it off (it has fallen to $0.23 < 0.5$). In between, at the same $h^{\text{ext}} = -0.5$, the neuron is high or low depending only on its history.

**THE STATE DEPENDS ON HISTORY**

At $t = 10\,\text{ms}$ and at $t = 60\,\text{ms}$ the input is the same, $h^{\text{ext}} = -0.5$, but the rate is 0 in one case and 0.98 in the other. The difference is what happened before. This dependence on past input (often called hysteresis) is what lets a single neuron with feedback hold information; the book develops it into a model of long-term memory in Ch. 5.

**PITFALL · THE PULSE MUST BE LONG ENOUGH**

Raising $h^{\text{ext}}$ is not enough by itself; the rate must cross the watershed 0.5 *before* the pulse ends. With the same parameters, a 5 ms pulse to $h^{\text{ext}} = 0$ brings $r$ only to 0.36. When $h^{\text{ext}}$ returns to $-0.5$, the rate is on the low side of 0.5 and decays back to 0: nothing is stored.

## > 10.8 The two ingredients: positive feedback and saturation

What makes bistability possible? We need an S-curve that can cross the identity line three times. Two things are required [LMLF 2.3.1]:

- **Positive feedback** that is strong enough: in the middle crossing the S-curve must be steeper than 1. Its maximal slope is $(\beta J)/4$ (the effective gain over 4), so we need

$$\boxed{\frac{\beta J}{4} > 1}.$$

  If this fails, the S-curve is flatter than the identity line everywhere and crosses it only once, whatever $h^{\text{ext}}$ is. If it holds, a suitable $h^{\text{ext}}$ produces three crossings.

- **Saturation**: the curve must bend back below the identity line at high $r$ (and stay above it at low $r$). That is what the bounded logistic function provides, and what the linear $g$ lacked: there, $J > 1$ meant an unstable fixed point and divergence, with no second stable state to land in.

**EXAMPLE · HOW MUCH FEEDBACK IS ENOUGH? ($\beta = 20$)**

$\beta J/4 > 1 \iff J > 4/\beta = 0.2$. With the book's $J = 1$: $\beta J/4 = 5$, comfortably bistable for some range of $h^{\text{ext}}$. With $J = 0.1$: $\beta J/4 = 0.5$, one fixed point for every $h^{\text{ext}}$, no memory.

**PITFALL · THREE INTERSECTIONS, NOT TWO**

The book says that the bistable regime has "two intersection points". It has *three* intersections (fixed points), of which *two* are stable. The unstable middle one is always there, between the two attractors.

## > 10.9 A technical aside: the case $\lambda = 1$

The linearisation decides stability whenever $\lambda \neq 1$. When $\lambda = 1$ exactly, it gives $\tau\dot{\delta r} = 0$, which says nothing. This happens precisely at the bifurcation points, where the S-curve is tangent to the identity line.

**Step 1: expand one order further** [LMLF eq. 2.23–2.25]. A second-order Taylor expansion adds the curvature term:

$$g(h^* + J\delta r) \approx g(h^*) + g'(h^*)J\delta r + \tfrac{1}{2}g''(h^*)J^2\delta r^2.$$

**Step 2: substitute**. The constant term vanishes (fixed point) and the linear term vanishes ($\lambda = 1$),

leaving

$$\tau\,\dot{\delta r} = \tfrac{1}{2}J^2 g''(h^*)\,\delta r^2.$$

**Step 3: read the sign.** $\delta r^2 > 0$ on *both* sides of $r^*$, so $\dot{\delta r}$ has the same sign (that of $g''$) on both sides. The flow goes towards $r^*$ on one side and away on the other.

DEF · HALF-STABLE FIXED POINT

A fixed point is half-stable if the dynamics approach it from one side and leave it on the other.

**Step 4: solve** (the equation is separable): $\dfrac{d\delta r}{\delta r^2} = c\,dt$ with $c = \frac{J^2 g''(h^*)}{2\tau}$, so $-\frac{1}{\delta r} = ct + \text{const}$ and

$$\delta r(t) = \frac{1}{1/\delta r(0) - c\,t}.$$

This is hyperbolic, not exponential: much slower near the fixed point.

EXAMPLE · AT THE BIFURCATION POINT $h^{\text{ext}} = -0.803$ ($\beta = 20$, $J = 1$, $\tau = 10\,\text{MS}$)

Here $r^* = 0.947$, and $g'' = \beta^2 g(1-g)(1-2g) = 400 \cdot 0.947 \cdot 0.053 \cdot (-0.894) = -17.9$, so $c = -17.9/20 = -0.89$ per ms.

- From above, $\delta r(0) = +0.01$: $\delta r(t) = 1/(100 + 0.89t)$. After 100 ms it is 0.0053: only halved. (An exponential decay with $\tau = 10$ ms would have shrunk it $e^{10} \approx 22\,000$-fold.)
- From below, $\delta r(0) = -0.01$: $\delta r(t) = 1/(-100 + 0.89t)$ grows in size and the approximation breaks down after about 100 ms; the neuron leaves towards the low state.

PITFALL · LINEARISATION TELLS YOU ONLY ABOUT THE NEIGHBOURHOOD

Both the first- and second-order expansions are valid only while $\delta r$ is small. They decide *local* stability. Whether a fixed point is reached from far away (global stability) needs the full picture, such as the vector field [LMLF 2.3.1]. And $\lambda = 1$ exactly is a knife-edge: the slightest change of $h^{\text{ext}}$ makes $\lambda \neq 1$, and the exponential behaviour takes over again.

PART B · BONUS HOUR 2

# Linear Algebra as Geometry

**What the bonus hour does.** It builds a geometric picture of matrices and eigenvectors, and then uses it on the lecture's network: linearize around a fixed point, get $\dot{\delta r} = M\delta r$, and decouple it by moving to the eigenvectors. It ends with shortcuts for finding eigenvectors without computing a characteristic polynomial. The bonus-hour notes have several slips (a missing minus sign, wrong factors in the final solution, a wrong eigenvalue claim). Each is corrected in a pitfall box.

SECTION 11

## Three ways to look at a matrix

A matrix $M$ (and a vector $\vec{v}$) can be seen as

1. a mathematical object with rules (how to add and multiply);
2. a compact way to write many equations at once: $M\vec{v} = \vec{b}$ is one line instead of $n$;
3. **a geometric action**: $\vec{v}$ is an arrow from the origin, and $M$ moves every arrow of the plane to a new arrow $M\vec{v}$.

This hour uses the third view.

SECTION 12

## A matrix moves the plane

EXAMPLE · A DIAGONAL MATRIX: STRETCH EACH AXIS

$M = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, $\vec{v} = (1,1)$: $M\vec{v} = (3,2)$. Split $\vec{v}$ into its two axis parts: $M(1,0) = (3,0)$ and $M(0,1) = (0,2)$. So $M$ stretches the first axis by 3 and the second by 2, and nothing else.

EXAMPLE · A NON-DIAGONAL MATRIX: THE AXES GET TILTED

$M = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$: $M(1,1) = (5,2)$, $M(1,0) = (3,1)$, $M(0,1) = (2,1)$. The columns of $M$ are where the axis vectors land: $e_1 \mapsto (3,1)$, $e_2 \mapsto (2,1)$. Both arrows are turned, so "stretch by how much, in which direction" is not visible from the axes.

PITFALL · "$\vec{v}$ GOT STRETCHED BY 5 IN ONE DIRECTION AND BY 2 IN ANOTHER"

The notes read $M(1,1) = (5,2)$ as "stretched by 5 and by 2". Those are only the *coordinates* of the new arrow, not stretch factors. For this $M$ the real stretch directions are its eigenvectors, with eigenvalues $2 \pm \sqrt{3} \approx 3.73$ and $0.27$: it stretches by 3.73 along one direction and *shrinks* to 0.27 along another. Its determinant is $3 \cdot 1 - 2 \cdot 1 = 1$, so the area does not change at all.

PITFALL · THE THIRD EXAMPLE OF THE NOTES

For $M = \begin{pmatrix} \frac{1}{2} & 1 \\ 2 & \frac{1}{10} \end{pmatrix}$ the notes say "$M$ stretches the vectors in one direction and shrinks them in another". But $e_1 \mapsto (\frac{1}{2}, 2)$ (length 2.06) and $e_2 \mapsto (1, \frac{1}{10})$ (length 1.005) both get longer. The eigenvalues are $0.3 \pm \sqrt{2.04} \approx 1.73$ and $-1.13$: both have $|\lambda| > 1$, so $M$ *stretches* along both eigendirections (and flips one of them, since $-1.13 < 0$). $\det M = 0.05 - 2 = -1.95$: the area grows by 1.95 and the orientation flips.

SECTION 13

# The determinant: how much area changes

DETERMINANT AS AN AREA FACTOR

$M$ maps the unit square to a parallelogram of area $|\det M|$. So $|\det M|$ is the factor by which *every* area is multiplied.

- $|\det M| > 1$: areas grow. $|\det M| < 1$: areas shrink.
- $\det M < 0$: the plane is also flipped (orientation reversed, like a mirror).
- $\det M = 0$: the plane is collapsed onto a line (or a point); areas become 0, and $M$ has no inverse. (This answers the notes' question.)

Examples: $\mathrm{Diag}(3,2)$: 6. $\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$: 1. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: 0, every vector lands on the line $x_1 = x_2$.

PITFALL · AREA, NOT EVERY DIRECTION

"$|\det M| > 1$ expands the plane" means the *area* grows. Single directions can still shrink: $\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$ keeps the area ($\det = 1$) while shrinking one direction to 0.27 of its length and stretching another by 3.73. The notes' link to the change-of-variables formula is right: in $\int f(x)\,dx$ with $x = Mu$, the factor $|\det M|$ is exactly this area (volume) factor.

SECTION 14

# Eigenvectors: the directions a matrix only stretches

**DEF · EIGENVECTOR, EIGENVALUE (RECAP OF WEEK 1)**

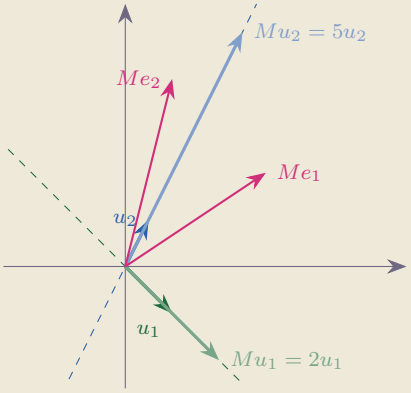
$\vec{u} \neq \vec{0}$ is an eigenvector of $M$ with eigenvalue $\lambda$ if $M\vec{u} = \lambda\vec{u}$: the arrow keeps its direction and is only multiplied by $\lambda$ (reversed if $\lambda < 0$).

**EXAMPLE · $M = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$**

The axes are tilted: $e_1 \mapsto (3, 2)$, $e_2 \mapsto (1, 4)$. Now try two special arrows:

$$M\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3-1 \\ 2-4 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad M\begin{pmatrix} 1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3/2+1 \\ 1+4 \end{pmatrix} = \begin{pmatrix} 5/2 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 1/2 \\ 1 \end{pmatrix}.$$

Along $(1, -1)$, $M$ stretches by 2; along $(\frac{1}{2}, 1)$, by 5. Check: $\det M = 12 - 2 = 10 = 2 \cdot 5$ and $\operatorname{Tr} M = 7 = 2 + 5$. ✓




$M = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$. The axis vectors are turned (pink). The eigenvectors stay on their own lines (dashed) and are only stretched, by 2 and 5.

**THE TAKE-HOME MESSAGE**

In the standard coordinates $M$ looks complicated. In the coordinates of its eigenvectors it is just "stretch by 2 along $u_1$, stretch by 5 along $u_2$": a **diagonal** matrix, like the first example. Choosing the eigenvectors as axes (the eigenbasis) is exactly Week 1's diagonalization $M = UDU^{-1}$.

SECTION 15

# From the network to $\dot{\delta r} = M\delta r$

The lecture's network of $N$ neurons, all inhibiting each other with strength $J > 0$ (with $\tau = 1$):

$$\dot{r}_i = -r_i + g\Big(-J\sum_{j=1}^{N} r_j + h_0\Big), \qquad i = 1, \ldots, N.$$

In one dimension we could draw $\dot{r}$ against $r$. In $N$ dimensions we cannot, so we **linearize**: look at a small deviation $\delta r_i = r_i - r_i^*$ from a fixed point $r^*$.

**Step by step.**

1. At the fixed point, every neuron has the same input $h^* = -J\sum_j r_j^* + h_0$, and $r_i^* = g(h^*)$.
2. Near it, the input is $h^* - J\sum_j \delta r_j$. First-order Taylor: $g(h^* - J\sum_j \delta r_j) \approx g(h^*) - Jg'(h^*)\sum_j \delta r_j$.
3. Subtract the fixed-point equation: $\dot{\delta r}_i = -\delta r_i - Jg'(h^*)\sum_{j=1}^N \delta r_j$.
4. Write $\sum_j \delta r_j = \mathbf{1}^T \delta r$, with $\mathbf{1} = (1, \ldots, 1)$. The vector whose every entry is $\sum_j \delta r_j$ is $\mathbf{1}\mathbf{1}^T \delta r$ (every row of $\mathbf{1}\mathbf{1}^T$ is all ones).

$$\boxed{\dot{\delta r} = M\delta r, \qquad M = -I - Jg'(h^*)\,\mathbf{1}\mathbf{1}^T}$$

**PITFALL · A LOST MINUS SIGN**

The notes write $\dot{\delta r}_i = -\delta r_i + Jg'(h^*)\sum_j \delta r_j$ and $M = -I + Jg'(h^*)\mathbf{1}\mathbf{1}^T$. The input is $-J\sum_j r_j$, so its derivative carries $-J$: **the correct sign is minus**. It matters: with $+$ the network would be excitatory and could become unstable; with $-$ (inhibition) every eigenvalue here is negative (§17). Also, the notes call the $\delta r_i$ "linearly independent vectors". They are numbers, the *coordinates* of the vector $\delta r$; the independent vectors are the axis vectors $e_i$.

SECTION 16

# Two inhibitory neurons, solved by decoupling

The 2-neuron example of the bonus hour (each neuron inhibits the *other* with strength $a$, no self-connection):

$$\dot{r}_1 = -r_1 - ar_2, \qquad \dot{r}_2 = -r_2 - ar_1, \qquad \text{i.e.} \quad \dot{\vec{r}} = M\vec{r}, \quad M = \begin{pmatrix} -1 & -a \\ -a & -1 \end{pmatrix}.$$

The equations are **coupled**: $\dot{r}_1$ depends on $r_2$ and vice versa, so we cannot solve each one alone.

**Step 1: the sum.** Let $r_+ = r_1 + r_2$. Then $\dot{r}_+ = (-r_1 - ar_2) + (-r_2 - ar_1) = -(1+a)(r_1 + r_2)$:

$$\dot{r}_+ = -(1+a)\, r_+ \quad \Rightarrow \quad r_+(t) = r_+(0)\, e^{-(1+a)t}.$$

**Step 2: the difference.** Let $r_- = r_1 - r_2$. Then $\dot{r}_- = (-r_1 - ar_2) - (-r_2 - ar_1) = -(r_1 - r_2) + a(r_1 - r_2)$:

$$\dot{r}_- = -(1-a)\, r_- \quad \Rightarrow \quad r_-(t) = r_-(0)\, e^{-(1-a)t}.$$

**Step 3: back to the neurons.** $r_+ + r_- = 2r_1$ and $r_+ - r_- = 2r_2$, so

$$\boxed{r_1(t) = \tfrac{1}{2}\left[r_+(0)e^{-(1+a)t} + r_-(0)e^{-(1-a)t}\right], \qquad r_2(t) = \tfrac{1}{2}\left[r_+(0)e^{-(1+a)t} - r_-(0)e^{-(1-a)t}\right]}$$

**PITFALL · THE NOTES' FINAL FORMULAS**

The notes write $r_+ + r_- = r_1$ and $r_- - r_+ = r_2$, and get $r_1 = e^{-(1-a)t} + e^{-(1+a)t}$, $r_2 = e^{-(1-a)t} - e^{-(1+a)t}$. Three problems:

- $r_+ + r_- = 2r_1$ and $r_+ - r_- = 2r_2$: a factor $\frac{1}{2}$ is missing, and the sign of $r_2$ is reversed.
- The constants $r_\pm(0)$ are missing: the notes silently took $r_+(0) = r_-(0) = 1$, i.e. $r_1(0) = 1$, $r_2(0) = 0$.
- Check their $r_1$ at $t = 0$: $1 + 1 = 2 \neq r_1(0)$. With the corrected formula and $r_1(0) = 1, r_2(0) = 0$: $r_1(0) = \frac{1}{2}(1 + 1) = 1$, $r_2(0) = 0$. ✓

**EXAMPLE · WITH NUMBERS: $a = 0.5$, $r_1(0) = 1$, $r_2(0) = 0$**

$r_+(0) = r_-(0) = 1$. At $t = 1$: $r_+ = e^{-1.5} = 0.223$, $r_- = e^{-0.5} = 0.607$, so $r_1 = \frac{1}{2}(0.223 + 0.607) = 0.415$ and $r_2 = \frac{1}{2}(0.223 - 0.607) = -0.192$. ($r$ here is a deviation from the fixed point, so negative values are allowed.) Both modes decay; the sum decays faster ($e^{-1.5t}$) than the difference ($e^{-0.5t}$).

**WHAT $r_\pm$ ARE: THE EIGENVECTORS**

$r_+$ measures the state along $(1, 1)$, and $r_-$ along $(1, -1)$. Indeed $M(1, 1) = -(1 + a)(1, 1)$ and $M(1, -1) = -(1 - a)(1, -1)$: these are the eigenvectors, with eigenvalues $-(1 + a)$ and $-(1 - a)$. Decoupling "by hand" was diagonalization all along. And it tells us the physics:

- the **common mode** $(1, 1)$ (both neurons together) always decays;
- the **difference mode** $(1, -1)$ (one up, one down) decays only if $a < 1$. If $a > 1$ it grows: one neuron wins and silences the other (winner-take-all).

In experiments we record $r_1$ and $r_2$, the neurons, so at the end we always translate back from $r_\pm$.

SECTION 17

# Shortcuts for eigenvectors

When a matrix has a pattern, its eigenvectors have a pattern too. Three examples.

**1. A MATRIX OF ALL ONES: $\mathbf{1}\mathbf{1}^T$ ($N \times N$)**

$(\mathbf{1}\mathbf{1}^T\vec{v})_j = \sum_i v_i$ for every $j$: every entry of the result is the sum of the entries of $\vec{v}$. So $\mathbf{1}\mathbf{1}^T\vec{v} = \lambda\vec{v}$ means $\sum_i v_i = \lambda v_j$ for every $j$. Two cases:

- all $v_j$ equal: $\vec{v} = \mathbf{1}$, and $\lambda = N$ (the sum of $N$ ones);
- $\sum_i v_i = 0$: then $\lambda = 0$. There are $N - 1$ independent such vectors, e.g. $(1, -1, 0, \dots)$, $(1, 0, -1, 0, \dots)$.

Eigenvalues: $N$ (once) and 0 ($N - 1$ times). Check: the trace of $\mathbf{1}\mathbf{1}^T$ is $N = N + 0 + \cdots + 0$. ✓

**PITFALL · THE NOTES' VERSION**

The notes conclude that "the sum of all coordinates of $\vec{v}$ has to be that eigenvalue". The condition is $\sum_i v_i = \lambda v_j$ for every $j$, not $\sum_i v_i = \lambda$. It gives exactly the two cases above.

**2. EQUAL ROW SUMS**

If every row of $M$ adds up to the same number $s$, then $M\mathbf{1} = s\mathbf{1}$: $\mathbf{1}$ is an eigenvector with eigenvalue $s$ (each entry of $M\mathbf{1}$ is a row sum). Example: $\begin{pmatrix} -1 & -a \\ -a & -1 \end{pmatrix}$, row sums $-1-a$: $\mathbf{1} = (1,1)$ with eigenvalue $-(1+a)$.

**3. SYMMETRIC MATRICES**

If $M = M^T$, eigenvectors with different eigenvalues are **orthogonal**: $\vec{u}^T\vec{v} = \sum_i u_i v_i = 0$. So once you know one eigenvector of a symmetric $2 \times 2$ matrix, the other is perpendicular to it: from $(1,1)$ we get $(1,-1)$, with eigenvalue $-(1-a)$ in the example above.

**EXAMPLE · ALL OF IT TOGETHER: $N$ NEURONS INHIBITING EACH OTHER**

Take $N$ neurons, each inhibiting every *other* neuron with effective strength $a = Jg'(h^*)$ and no self-connection. Then $M = -I - a(\mathbf{1}\mathbf{1}^T - I)$. Using shortcut 1:

- along $\mathbf{1}$: $-1 - a(N-1)$ (the common mode, always decaying);
- along any $\vec{v}$ with $\sum v_i = 0$: $-1 - a(0-1) = -1 + a$ (the difference modes).

Stable if and only if $a < 1$. For $N = 2$ this gives $-(1+a)$ and $-(1-a)$, as before. ✓(If each neuron also inhibits itself, as in the lecture's $\sum_{j=1}^{N}$, then $M = -I - a\mathbf{1}\mathbf{1}^T$, with eigenvalues $-1 - aN$ and $-1$: always stable.)

PART C · TA SESSION 2

# Exam 2015, Question 2: Excitation and Inhibition

**What the TA session does.** It solves a past exam question: fixed points and stability for (1) one excitatory population with self-excitation and (2) an excitatory and an inhibitory population coupled together, with tanh as the activation function. It is the Part A analysis (fixed points, linearization, slope vs. 1) plus Week 1's $2 \times 2$ trace–determinant test. The session stopped after part 1.2.2; that part is completed here. Slips in the TA notes are corrected in pitfall boxes.

SECTION 18

## The model and the function tanh

Each population $i$ has activity $S_i(t)$ and

$$\dot{S}_i = -S_i + \tanh(h_i), \qquad h_i = \sum_j J_{ij} S_j,$$

where $J_{ij}$ is the connection *from* population $j$ *to* $i$ (time measured in units of $\tau$).

DEF · HYPERBOLIC TANGENT

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

WHAT WE NEED ABOUT tanh

- It is an S-curve from $-1$ to 1: $\tanh 0 = 0$, $\tanh x \to \pm 1$ as $x \to \pm\infty$. It is **odd**: $\tanh(-x) = -\tanh x$.
- Derivative: $\tanh' x = 1 - \tanh^2 x$. So $\tanh' 0 = 1$: near 0, $\tanh x \approx x$.
- For $x > 0$: $\tanh x < x$, and tanh is concave (bends down).

Numbers: $\tanh 0.5 = 0.462$, $\tanh 1 = 0.762$, $\tanh 2 = 0.964$.

**Why** $\tanh' = 1 - \tanh^2$**:** quotient rule with $\sinh' = \cosh$, $\cosh' = \sinh$: $\tanh' = \dfrac{\cosh^2 - \sinh^2}{\cosh^2} = 1 - \tanh^2$.

tanh VS. THE LOGISTIC FUNCTION OF PART A

$\tanh x = 2\sigma(2x) - 1$, where $\sigma(x) = 1/(1 + e^{-x})$: the same S-shape, shifted and rescaled to the range $(-1, 1)$. Because $S$ can be negative, it is best read as a *deviation* from a baseline rate. (The TA's remark: today one prefers non-negative activation functions, since a firing rate

cannot be negative.)

SECTION 19

# Question 1.1: one excitatory population

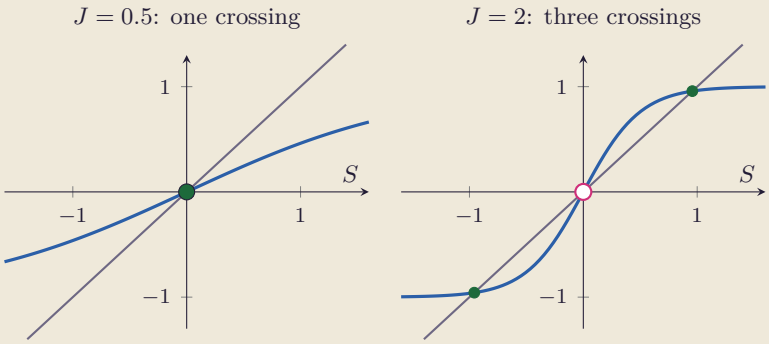



One excitatory population exciting itself with strength $J > 0$ (the TA's Figure 1).

**Question.** $\dot{S} = -S + \tanh(JS)$, $J > 0$. Find the fixed points and their stability as a function of $J$.

## > 19.1 Fixed points

1. Fixed point: $\dot{S} = 0 \iff S = \tanh(JS)$. There is no closed-form solution, so **draw both sides**: the identity line $y = S$ and the S-curve $y = \tanh(JS)$. Every crossing is a fixed point.
2. $S^* = 0$ always works: $\tanh(J \cdot 0) = 0$.
3. The S-curve's slope at 0 is $J \tanh'(0) = J$.
   - $J \leq 1$: for $S > 0$, $\tanh(JS) < JS \leq S$, so the curve is below the line: no other crossing. **One fixed point**, $S^* = 0$.
   - $J > 1$: just right of 0 the curve is above the line (slope $J > 1$), but it never exceeds 1 while the line keeps rising. So they cross exactly once more, at some $S^* \in (0, 1)$; by oddness also at $-S^*$. **Three fixed points**: $-S^*, 0, S^*$.




Grey: $y = S$. Blue: $y = \tanh(JS)$. Filled green = stable, open pink = unstable (shown below). For $J = 2$ the outer crossings are at $\pm 0.9575$.

## > 19.2 Stability, step by step

1. Perturb: $S = S^* + \delta S$. Then $\dot{\delta S} = -(S^* + \delta S) + \tanh(JS^* + J\delta S)$.
2. First-order Taylor: $\tanh(JS^* + J\delta S) \approx \tanh(JS^*) + J(1 - \tanh^2(JS^*))\delta S$.

3. $-S^* + \tanh(JS^*) = 0$ (fixed point), so $\dot{\delta S} = \lambda\, \delta S$ with

$$\lambda = -1 + J(1 - \tanh^2(JS^*)) = \boxed{-1 + J(1 - S^{*2})},$$

using $\tanh(JS^*) = S^*$ once more.

4. $\delta S(t) = \delta S(0) e^{\lambda t}$: **stable if** $\lambda < 0$, unstable if $\lambda > 0$. Equivalently: the S-curve's slope at the crossing, $J(1 - S^{*2})$, must be **less than 1**, the slope of the identity line (the same rule as Part A, §10).

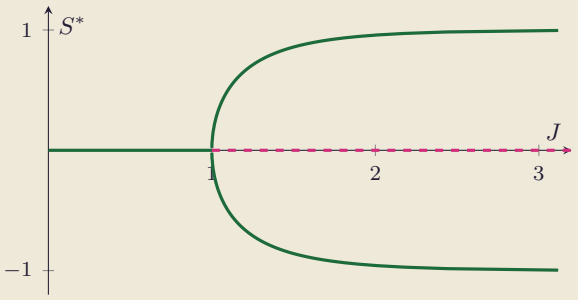
**Reading off the cases.**

- $S^* = 0$: $\lambda = J - 1$. Stable for $J < 1$, **unstable for** $J > 1$.
- $S^* = \pm S^* \neq 0$ (only for $J > 1$): $\lambda = -1 + J(1 - S^{*2}) < 0$, **stable**. (Proof in the box below; the two have the same $\lambda$ because $S^{*2}$ is the same.)
- $J = 1$: $\lambda = 0$ at $S^* = 0$, so the linearization is silent. But $\tanh S < S$ for $S > 0$, so $\dot{S} = -S + \tanh S < 0$ for $S > 0$ and $> 0$ for $S < 0$: the flow points to 0 from both sides. Stable, though slowly ($\dot{S} \approx -S^3/3$).

| $J$ | $S^* > 0$ | SLOPE $J(1 - S^{*2})$ | $\lambda$ |
|---|---|---|---|
| 1.2 | 0.659 | 0.680 | −0.32 |
| 1.5 | 0.859 | 0.394 | −0.61 |
| 2 | 0.958 | 0.166 | −0.83 |
| 3 | 0.995 | 0.031 | −0.97 |

**WHY THE OUTER FIXED POINTS ARE ALWAYS STABLE**

Let $u = JS^* > 0$. The fixed-point equation says $\tanh u = S^*$, so $\dfrac{\tanh u}{u} = \dfrac{1}{J}$. We need $J(1 - \tanh^2 u) < 1$, i.e. $1 - \tanh^2 u < \dfrac{\tanh u}{u}$. Multiply by $u \cosh^2 u > 0$: this is $u < \sinh u \cosh u = \frac{1}{2} \sinh 2u$, i.e. $\sinh 2u > 2u$, which holds for every $u > 0$ ($\sinh y > y$ for $y > 0$). ✓




Fixed points against $J$ (a pitchfork bifurcation). Solid = stable, dashed = unstable. At $J = 1$ the single stable state 0 becomes unstable and two new stable states $\pm S^*$ appear.

THE SAME LESSON AS PART A

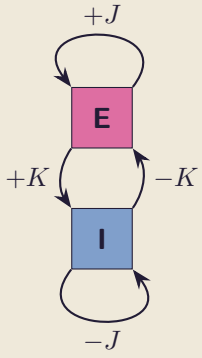
Strong enough positive feedback ($J > 1$) plus saturation gives **two stable states**: the population can sit "high" or "low" and stays there. That is bistability, a memory, exactly as for the logistic autapse.

PITFALL · CORRECTIONS TO THE TA NOTES, 1.1

- "$\tanh^2(S) > 0$, so $1 - \tanh^2(S) < 1$ always": $\tanh^2(0) = 0$, so it is $\geq 0$ and $1 - \tanh^2 \leq 1$.
- "$-1 + J > -1 + 1 > 0$": should read $-1 + J > -1 + 1 = 0$.
- The comparison "$1 \lessgtr J(1 - \tanh^2(0))$" for the *outer* fixed points should be at $JS^*$: $1 \lessgtr J(1 - \tanh^2(JS^*))$. And it compares the slope of $S$ with the slope of $\tanh(JS)$, not of "$\tanh^2(JS)$".
- The conclusion (outer points stable) is right, but the notes only read it off a picture. The box above proves it.

SECTION 20

# Question 1.2: an excitatory and an inhibitory population




The TA's Figure 2 ($J, K > 0$): E excites itself ($+J$) and I ($+K$); I inhibits itself ($-J$) and E ($-K$).

Reading the input from the figure ("from $j$ to $i$"):

$$\dot{S}_E = -S_E + \tanh(JS_E - KS_I), \qquad \dot{S}_I = -S_I + \tanh(KS_E - JS_I).$$

### 1.2.1: $(0,0)$ is a fixed point; its stability

1. At $(0,0)$ both inputs are 0 and $\tanh 0 = 0$: $\dot{S}_E = \dot{S}_I = 0$. ✓
2. Linearize. Near 0, $\tanh x \approx x$ (slope 1), so

$$\delta\dot{S}_E \approx -\delta S_E + J\delta S_E - K\delta S_I, \qquad \delta\dot{S}_I \approx -\delta S_I + K\delta S_E - J\delta S_I.$$

(The same as taking the four partial derivatives, e.g. $\partial\dot{S}_E/\partial S_E = -1 + J(1 - \tanh^2(\cdot))$, and setting $S_E = S_I = 0$.)

3. In matrix form:
$$\begin{pmatrix}\delta\dot{S}_E\\ \delta\dot{S}_I\end{pmatrix} = A\begin{pmatrix}\delta S_E\\ \delta S_I\end{pmatrix}, \qquad A = \begin{pmatrix}J-1 & -K\\ K & -J-1\end{pmatrix}.$$

4. Week 1's $2\times 2$ test: $\operatorname{Tr} A = (J-1)+(-J-1) = -2 < 0$ always, and $\det A = (J-1)(-J-1) - (-K)(K) = 1 - J^2 + K^2$.

5. Stable $\iff$ $\operatorname{Tr} < 0$ and $\det > 0$ $\iff$ $K^2+1 > J^2$ $\iff$ $\boxed{J < \sqrt{K^2+1}}$ (as $J > 0$).

**The eigenvalues themselves** (useful for 1.2.2). $\lambda^2 - \operatorname{Tr}\,\lambda + \det = \lambda^2 + 2\lambda + (1 - J^2 + K^2) = 0$, so

$$\boxed{\lambda_\pm = -1 \pm \sqrt{J^2 - K^2}}$$

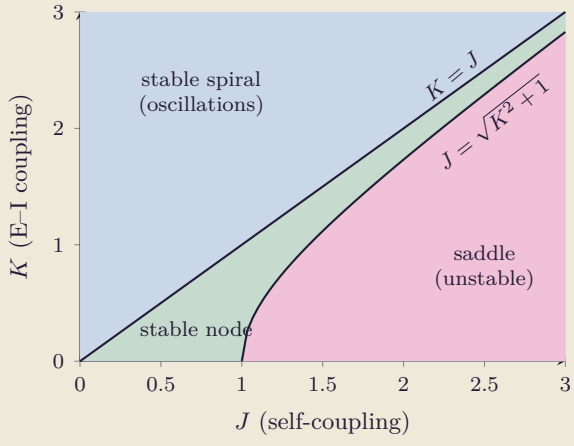
## 1.2.2: the qualitatively different behaviours

**DEF · COMPLEX EIGENVALUES MEAN OSCILLATIONS**

If $\lambda = \alpha \pm i\omega$, the solutions contain $e^{\lambda t} = e^{\alpha t}(\cos\omega t \pm i\sin\omega t)$: a rotation at angular frequency $\omega$ whose size changes like $e^{\alpha t}$. $\alpha < 0$: damped oscillation (the state spirals into the fixed point, a stable spiral). $\alpha > 0$: growing oscillation. The eigenvalues are complex exactly when $\operatorname{Tr}^2 - 4\det < 0$.

Here $\operatorname{Tr}^2 - 4\det = 4 - 4(1 - J^2 + K^2) = 4(J^2 - K^2)$. So everything depends on comparing $J$ with $K$, and $J$ with $\sqrt{K^2+1}$:

| REGIME | EIGENVALUES | BEHAVIOUR NEAR $(0,0)$ |
|---|---|---|
| $J < K$ | $-1 \pm i\sqrt{K^2 - J^2}$ | stable spiral: **damped oscillations**, frequency $\sqrt{K^2-J^2}$, decay $e^{-t}$ |
| $J = K$ | $-1$ (twice) | stable, decays like $e^{-t}$, no oscillation |
| $K < J < \sqrt{K^2+1}$ | two real, both $< 0$ | stable node: decays without oscillating |
| $J = \sqrt{K^2+1}$ | $0$ and $-2$ | borderline: one decaying direction, one neutral; linearization undecided |
| $J > \sqrt{K^2+1}$ | one $> 0$, one $< 0$ | saddle: **unstable**, activity runs away along one direction (until tanh saturates) |



The three regimes of $(0,0)$ in the $(J,K)$ plane. Above $K=J$: oscillations. Between the two curves: plain decay. Right of $J=\sqrt{K^2+1}$: unstable.

**EXAMPLE · ONE POINT IN EACH REGION**

- $J=1$, $K=2$: $\lambda=-1\pm i\sqrt{3}$. Damped oscillation with period $2\pi/\sqrt{3}\approx 3.6$ time units.
- $J=1.2$, $K=1$: $\lambda=-1\pm\sqrt{0.44}=-0.34,\,-1.66$ (check: $\det=1-1.44+1=0.56=0.34\cdot 1.66$). Stable node.
- $J=2$, $K=1$: $\lambda=-1\pm\sqrt{3}=0.73,\,-2.73$ ($\det=-2<0$). Saddle: unstable.

**WHY THE E-I LOOP OSCILLATES**

E excites I, and I then inhibits E; with E low, I loses its drive and falls, which releases E, and so on. That is a rotation in the $(S_E,S_I)$ plane. The loop strength $K$ drives the rotation; the self-coupling $J$ pushes the two populations apart in opposite directions (E amplifies itself, I damps itself). When $K>J$ the rotation wins: oscillations. When $J$ is too large compared with $K$, self-excitation of E wins: instability.

**PITFALL · CORRECTIONS TO THE TA NOTES, 1.2**

- "We can check for oscillations by looking at $p-4q$": the discriminant is $p^2-4q$ (trace squared). Here $p^2-4q=4(J^2-K^2)$: oscillations iff $K>J$.
- "For the fixed point to be stable we need all eigenvalues to be negative": for complex eigenvalues, read "negative *real part*". Here the real part is $-1$ whenever $K>J$, so those cases are always stable.
- "If $J=\sqrt{K^2+1}$ we have one stable direction": true (eigenvalue $-2$), but the other eigenvalue is 0, so the linear analysis cannot decide stability there.
- The notes stop at 1.2.2 without listing the regimes; the table and the figure above complete it.

WEEK 02

# Practice Problems and Summary

SECTION 21

## Practice problems with full solutions

**EXAMPLE · P1: FIXED POINT AND STABILITY**

$\tau\dot{r} = -r + 0.8r + 3$. Find $r^*$, decide stability, and give $r(t)$ for $r(0) = 0$, $\tau = 10\,\text{ms}$.

**Solution.** $J = 0.8$, $h^{\text{ext}} = 3$: $r^* = 3/0.2 = 15$. $J < 1$, so stable. $r(t) = 15(1 - e^{-0.2t/10}) = 15(1 - e^{-t/50})$: the time scale is 50 ms, five times $\tau$.

**EXAMPLE · P2: WHICH $J$ GIVES A TARGET RATE?**

With $h^{\text{ext}} = 2$, what $J$ makes the stable fixed point $r^* = 8$?

**Solution.** $8 = 2/(1 - J) \Rightarrow 1 - J = 0.25 \Rightarrow J = 0.75$. Since $0.75 < 1$ it is stable. ✓

**EXAMPLE · P3: PERTURBATION BY HAND**

$J = 1.2$, $h^{\text{ext}} = -0.4$, $\tau = 10\,\text{ms}$. Find $r^*$. The rate starts at $r(0) = 2.1$; what is it after 50 ms?

**Solution.** $r^* = -0.4/(1 - 1.2) = -0.4/(-0.2) = 2$. $J > 1$: unstable. $\delta r(0) = 0.1$, and $\delta r(t) = 0.1\, e^{0.2t/10} = 0.1\, e^{t/50}$. At $t = 50$: $\delta r = 0.1e \approx 0.27$, so $r \approx 2.27$. The rate drifts away from 2, slowly at first and then faster and faster.

**EXAMPLE · P4: VECTOR FIELD**

Sketch $\tau\dot{r}$ against $r$ for $\tau\dot{r} = -0.5r + 1$ and use it to say what happens from $r(0) = 5$.

**Solution.** A line with slope $-0.5$ crossing zero at $r^* = 2$. At $r = 5$: $\tau\dot{r} = -1.5 < 0$, so $r$ decreases, and it keeps decreasing until it reaches 2. The arrows point towards 2 from both sides: stable.

**EXAMPLE · P5: THE BORDER**

What happens for $J = 1$, $h^{\text{ext}} = 0.5$, $\tau = 10\,\text{ms}$, $r(0) = 0$?

**Solution.** $\tau\dot{r} = 0.5$, so $\dot{r} = 0.05$ per ms: $r(t) = 0.05t$, a straight line. There is no fixed point; the rate grows steadily, not exponentially.

## The non-linear autapse

**EXAMPLE · P6: READING A LOGISTIC FUNCTION**

$g(x) = 1/(1 + e^{-10(x-0.2)})$. Find $g(0.2)$, $g(0.4)$, $g(0)$ and the maximal slope.

**Solution.** $\theta = 0.2$, $\beta = 10$. $g(0.2) = 0.5$ (the threshold). $g(0.4) = 1/(1 + e^{-2}) = 0.881$. $g(0) = 1/(1 + e^{2}) = 0.119$ (symmetric about the threshold). Maximal slope $\beta/4 = 2.5$, at $x = 0.2$.

**EXAMPLE · P7: EFFECTIVE THRESHOLD**

$\theta = 0$, $J = 2$. Where along the $r$ axis is the midpoint of the S-curve $g(Jr + h^{\text{ext}})$ for $h^{\text{ext}} = -1$? For $h^{\text{ext}} = -0.4$? Which way did it move?

**Solution.** Midpoint $(\theta - h^{\text{ext}})/J$: $(0 + 1)/2 = 0.5$, and $(0 + 0.4)/2 = 0.2$. Raising $h^{\text{ext}}$ moved it *left*, from 0.5 to 0.2.

**EXAMPLE · P8: STABILITY BY $\lambda$**

$\theta = 0$, $\beta = 20$, $J = 1$, $h^{\text{ext}} = -0.3$. One fixed point is $r^* = 0.243$; check that it is one, and decide its stability. What about $r^* \approx 0.0026$?

**Solution.** $g(0.243 - 0.3) = g(-0.057) = 1/(1 + e^{1.14}) = 0.242 \approx 0.243$ ✓. $\lambda = 20 \cdot 0.243 \cdot 0.757 = 3.68 > 1$: unstable. For $r^* = 0.0026$: $\lambda = 20 \cdot 0.0026 \cdot 0.9974 = 0.052 < 1$: stable. (The third fixed point, $r^* \approx 1$, is also stable: $h^{\text{ext}} = -0.3$ is inside the bistable band.)

**EXAMPLE · P9: CAN IT BE BISTABLE AT ALL?**

(a) $\beta = 2$, $J = 1$. (b) $\beta = 20$, $J = 0.15$. (c) $\beta = 8$, $J = 1$. For which can a suitable $h^{\text{ext}}$ give two stable states?

**Solution.** Test $\beta J/4 > 1$: (a) 0.5, no. (b) 0.75, no. (c) 2, yes. In (a) and (b) the S-curve is everywhere flatter than the identity line, so it crosses it once whatever $h^{\text{ext}}$ is.

**EXAMPLE · P10: LINEAR AS A SPECIAL CASE**

Show that the stability rule $\lambda < 1$ reproduces the linear autapse result.

**Solution.** $g(h) = h$ has $g' = 1$ everywhere, so $\lambda = J \cdot 1 = J$. The rule becomes $J < 1$: stable, $J > 1$: unstable, exactly as in the first half of the week.

**EXAMPLE · P11: WRITING A BIT**

With $\theta = 0$, $\beta = 20$, $J = 1$, $\tau = 10\,\text{ms}$, the neuron rests at the low state with $h^{\text{ext}} = -0.5$. You can pulse $h^{\text{ext}}$ to 0. A 5 ms pulse brings $r$ to 0.36, a 15 ms pulse to 0.76. Which pulse stores a "1"? Why does setting $h^{\text{ext}} = -0.3$ forever not count as memory?

**Solution.** Only the 15 ms pulse: 0.76 is above the unstable fixed point 0.5 of $h^{\text{ext}} = -0.5$, so the rate continues to the high state; 0.36 is below it and decays back. Memory means the state persists *after* the input returns to its baseline; a permanent change of $h^{\text{ext}}$ is just a different input, not a stored bit.

**EXAMPLE · P12: HALF-STABLE**

At the bifurcation point $h^{\text{ext}} = -0.197$ ($\beta = 20$, $J = 1$), $r^* = 0.053$. Is $g''(h^*)$ positive or negative, and from which side is the fixed point approached?

**Solution.** $g'' = \beta^2 g(1-g)(1-2g)$ with $g = 0.053 < 0.5$, so $g'' > 0$. Then $\tau\dot{\delta r} = \frac{1}{2}J^2 g''\delta r^2 > 0$ on both sides: $r$ increases. From below ($\delta r < 0$) it moves up towards $r^*$; from above it moves away, up towards the high state. Half-stable, approached from the left.

## Bonus hour and TA session problems

**EXAMPLE · P-B1: EIGENVECTORS WITHOUT A POLYNOMIAL**

$M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Find its eigenvectors and eigenvalues with the shortcuts, and its area factor.

**Solution.** Equal row sums 3: $\mathbf{1} = (1,1)$ with $\lambda = 3$. Symmetric, so the other eigenvector is perpendicular: $(1,-1)$, and $M(1,-1) = (1,-1)$, so $\lambda = 1$. Area factor $|\det M| = 4-1 = 3 = 3\cdot 1$. ✓

**EXAMPLE · P-B2: THREE NEURONS INHIBITING EACH OTHER**

Three neurons, each inhibiting the other two with effective strength $a$ (no self-connection): $M = -I - a(\mathbf{1}\mathbf{1}^T - I)$. Find the eigenvalues for $a = 0.4$ and for $a = 1.5$. Stable?

**Solution.** Common mode $\mathbf{1}$: $-1-a(N-1) = -1-2a$. Difference modes (sum zero, twice): $-1+a$. $a = 0.4$: $-1.8$, $-0.6$, $-0.6$: stable. $a = 1.5$: $-4$, $0.5$, $0.5$: unstable; the difference modes grow, so some neurons win and silence the others.

**EXAMPLE · P-B3: WINNER-TAKE-ALL**

Two neurons, $\dot{r}_1 = -r_1 - 2r_2$, $\dot{r}_2 = -r_2 - 2r_1$, start at $r_1(0) = 1.1$, $r_2(0) = 0.9$. Find $r_1(t)$, $r_2(t)$ and describe them at $t = 2$.

**Solution.** $a = 2$. $r_+(0) = 2$, $r_-(0) = 0.2$. $r_+ = 2e^{-3t}$, $r_- = 0.2e^{t}$. $r_1 = \frac{1}{2}(2e^{-3t} + 0.2e^{t})$, $r_2 = \frac{1}{2}(2e^{-3t} - 0.2e^{t})$. At $t = 2$: $r_+ = 0.005$, $r_- = 1.478$, so $r_1 \approx 0.74$, $r_2 \approx -0.74$. A 10% head start of neuron 1 has grown: neuron 1 wins (in a real network, until saturation).

**EXAMPLE · P-C1: STABILITY OF AN OUTER FIXED POINT**

For $\dot{S} = -S + \tanh(1.5S)$, the positive fixed point is $S^* = 0.859$. Find $\lambda$ there and at $S^* = 0$.

**Solution.** $\lambda = -1 + J(1 - S^{*2}) = -1 + 1.5(1 - 0.738) = -1 + 0.394 = -0.61 < 0$: stable. At 0: $\lambda = J - 1 = 0.5 > 0$: unstable.

**EXAMPLE · P-C2: AN OSCILLATING E-I PAIR**

In Question 1.2 take $J = 0.5$, $K = 1$. Classify $(0,0)$ and give the oscillation period.

**Solution.** Tr $= -2$, det $= 1 - 0.25 + 1 = 1.75 > 0$: stable. $J < K$: $\lambda = -1 \pm i\sqrt{1 - 0.25} = -1 \pm 0.866i$, a stable spiral. Period $2\pi/0.866 \approx 7.3$ time units; the amplitude shrinks like $e^{-t}$, so only about one visible turn.

EXAMPLE · P-C3: RANGES

With $K = 1$ in Question 1.2, for which $J > 0$ is $(0, 0)$ stable, and for which does it oscillate?

**Solution.** Stable iff $J < \sqrt{K^2 + 1} = \sqrt{2} \approx 1.41$. Oscillating iff $J < K = 1$. So: $0 < J < 1$ stable spiral; $J = 1$ stable, no oscillation; $1 < J < 1.41$ stable node; $J > 1.41$ saddle.

SECTION 22

# Cheat sheet

WEEK 02 ON ONE PAGE

- **Autapse**: a neuron's synapse onto itself, strength $J$. $\tau\dot{r} = -r + g(Jr + h^{\text{ext}})$. Excitatory: $J > 0$ (positive feedback).
- **Linear** $g(h) = h$: $\tau\dot{r} = -(1 - J)r + h^{\text{ext}}$. Leak $-r$ vs. feedback $+Jr$.
- **Fixed point**: $\dot{r} = 0$. Linear autapse: $r^* = h^{\text{ext}}/(1 - J)$ $(J \neq 1)$. For $0 < J < 1$, an amplification by $1/(1 - J)$.
- **Stable**: nearby states return; **unstable**: nearby states run away. Only stable fixed points are seen in practice.
- **Perturbation**: $\delta r = r - r^*$; $\tau\dot{\delta r} = -(1 - J)\delta r$; $\delta r(t) = \delta r(0)e^{-(1-J)t/\tau}$. $h^{\text{ext}}$ drops out.
- $J < 1$: stable, decay time $\tau/(1 - J)$ (slower as $J \to 1$). $J > 1$: unstable, divergence. $J = 1$: border (no fixed point, or a whole line of them).
- **Full solution**: $r(t) = r^* + (r(0) - r^*)e^{-(1-J)t/\tau}$.
- **Vector field**: plot $\dot{r}$ vs. $r$; crossing zero going down = stable, going up = unstable.
- Stability = sign of the eigenvalue of the perturbation dynamics (here $-(1 - J)/\tau$), as in Week 1.

THE NON-LINEAR AUTAPSE ON ONE PAGE

- **Logistic** $g(x) = 1/(1 + e^{-\beta(x-\theta)})$: between 0 and 1, $g(\theta) = 0.5$, $g' = \beta g(1 - g)$, maximal slope $\beta/4$. $\beta \to \infty$: a step.
- **Fixed points**: $r^* = g(Jr^* + h^{\text{ext}})$, no closed form. Intersect the identity line with the S-curve $g(Jr + h^{\text{ext}})$ (effective gain $\beta J$, midpoint $(\theta - h^{\text{ext}})/J$; raising $h^{\text{ext}}$ slides it left).
- **Stability**: $\lambda = Jg'(Jr^* + h^{\text{ext}})$ = slope of the S-curve at the crossing. $\lambda < 1$ stable, $\lambda > 1$ unstable; $\delta r(t) = \delta r(0)e^{-(1-\lambda)t/\tau}$. Linear $g$: $\lambda = J$.
- **Global** stability: reached from every start. **Local**: reached from nearby starts. Linearisation only tests local stability.
- **Bifurcation**: qualitative change from a smooth parameter change. **Saddle-node**: a stable and an unstable fixed point merge and vanish; the curves are tangent $(\lambda = 1)$.
- Book values $\theta = 0$, $\beta = 20$, $J = 1$: three fixed points for $-0.803 < h^{\text{ext}} < -0.197$, otherwise one.

- **Attractor**: region all nearby trajectories go to. **Bistable**: two stable fixed points, an unstable one between them as the watershed.
- **Memory**: a brief pulse of $h^{\text{ext}}$ past a bifurcation point flips the state, which persists after the pulse (the state depends on history).
- **Ingredients**: positive feedback ($\beta J/4 > 1$) and saturation.
- $\lambda = 1$: second order, $\tau\dot{\delta r} = \frac{1}{2}J^2 g''\delta r^2$: **half-stable**, hyperbolic (slow) approach.

## BONUS HOUR 2 ON ONE PAGE

- A matrix moves the plane: its columns are the images of $e_1, e_2$. $|\det M|$ = area factor; $\det < 0$ flips; $\det = 0$ collapses onto a line.
- Eigenvector: $M\vec{u} = \lambda\vec{u}$, a direction only stretched. In the eigenbasis $M$ is diagonal ($M = UDU^{-1}$).
- Network linearization: $\dot{r}_i = -r_i + g(-J\sum_j r_j + h_0) \Rightarrow \dot{\delta r} = (-I - Jg'(h^*)\mathbf{1}\mathbf{1}^T)\delta r$ (minus sign!).
- Two mutually inhibiting neurons: $r_\pm = r_1 \pm r_2$, $\dot{r}_+ = -(1+a)r_+$, $\dot{r}_- = -(1-a)r_-$; $r_{1,2} = \frac{1}{2}(r_+ \pm r_-)$. $a > 1$: winner-take-all.
- $\mathbf{1}\mathbf{1}^T$: eigenvalue $N$ on $\mathbf{1}$, 0 on sum-zero vectors. Equal row sums $s$: $M\mathbf{1} = s\mathbf{1}$. Symmetric: orthogonal eigenvectors.

## TA SESSION 2 ON ONE PAGE

- tanh: odd, range $(-1, 1)$, $\tanh' = 1 - \tanh^2$, $\tanh' 0 = 1$, $\tanh x < x$ for $x > 0$.
- $\dot{S} = -S + \tanh(JS)$: $J \le 1$ one fixed point 0 (stable); $J > 1$ three: 0 unstable, $\pm S^*$ stable (pitchfork at $J = 1$).
- Stability: $\lambda = -1 + J(1 - S^{*2})$; stable iff the S-curve crosses the identity with slope $< 1$.
- E–I pair: $A = \begin{pmatrix} J-1 & -K \\ K & -J-1 \end{pmatrix}$, $\text{Tr} = -2$, $\det = 1 - J^2 + K^2$, $\lambda = -1 \pm \sqrt{J^2 - K^2}$.
- Stable iff $J < \sqrt{K^2+1}$. Oscillations iff $\text{Tr}^2 - 4\det < 0 \iff K > J$ (damped, frequency $\sqrt{K^2 - J^2}$). $J > \sqrt{K^2+1}$: saddle.